AECO Safety Vision – Helmet and Vest Detection

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- Computer Vision Prototype for Construction Safety Monitoring

Problem Statement

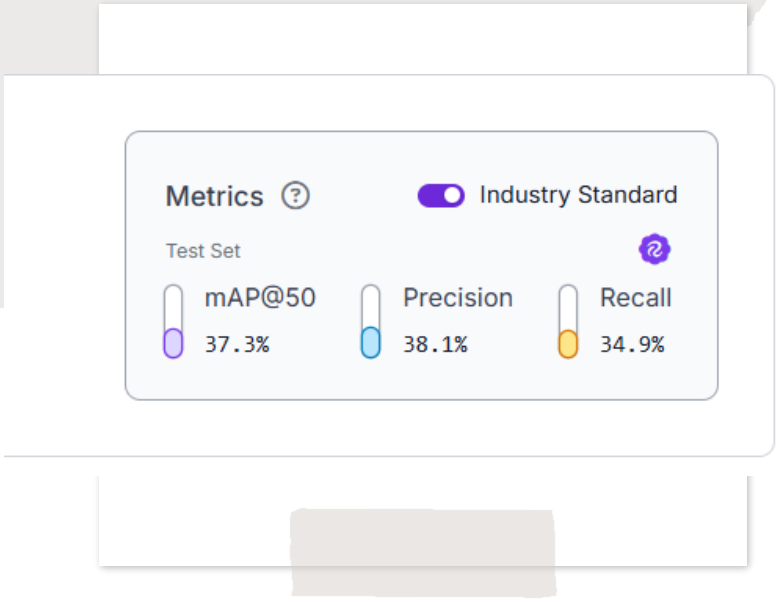
Construction sites present safety risks when workers do not wear required protective equipment.

Manual monitoring is difficult and time-consuming.

Computer vision can help detect safety equipment automatically.

Goal: Detect helmets and safety vests in construction site images.

Dataset



- Dataset created using Roboflow

- Total images: 57
- Train: 41
- Validation: 12
- Test: 4

- Classes:
- Helmet
- No-Helmet
- Vest
- No-Vest

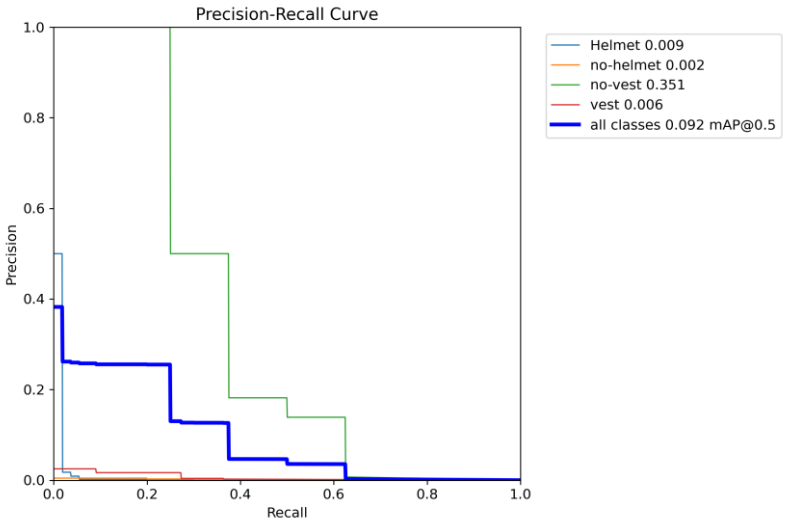
Model

- Model: YOLOv11 Object Detection (Fast)
- Training parameters:
- Epochs: 30
- Batch size: 16
- Image size: 640
- Training environment:
- Google Colab
- Ultralytics YOLO

Results

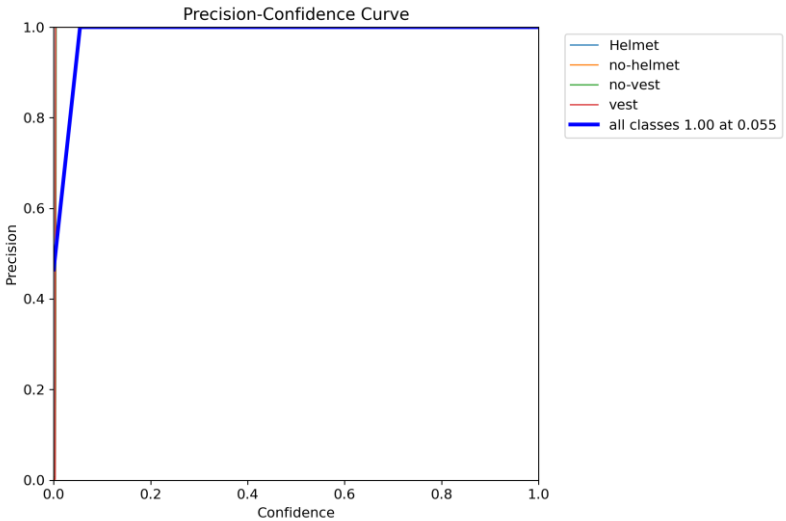
- Validation Metrics:
- Precision: 53.3%
- Recall: 27.6%
- mAP@50: 35.2%

- The model detects safety equipment but performance is limited by dataset size.



Limitations

- Model struggles with:
- - crowded construction scenes
- - very small helmets
- - strong lighting
- - selfie-style images



Conclusion

- Computer vision can assist safety monitoring in construction sites.
- The model could help identify workers missing helmets or safety vests.
- Future improvements:
- - Larger dataset
- - More diverse construction scenes
- - Better detection of small objects